Lecture 11: Duality (cont'd) and Rank

MATH 110-3

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Last time: Dual Space

Def'n:

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Prop'n:

For V finite-dimensional, V' is also finite-dimensional and $\dim V' = \dim V$.

Dual Basis

Def'n:

If $v_1, \ldots, v_n$ is a basis of V, then the dual basis of $v_1, \ldots, v_n$ is the list of $\phi_1, \ldots, \phi_n$ of V' such that

$$\phi_j(v_k) = \begin{cases} 1 & k = j, \\ 0 & k \neq j \end{cases}$$

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$$\frac{(x^k)^{(j)}(0)}{j!}$$

Last time: Dual Map

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Today we will see the following results:

Prop'ns:

Suppose V, W are finite dimensional and $T \in \mathcal{L}(V, W)$. Then

- T is surjective if and only if T' is injective
- T is injective if and only if T' is surjective

Examples and Exercises

Define $T : \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ by $(Tp)(x) = x^2 p(x)$ for $x \in \mathbb{R}$.

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Suppose $\phi \in \mathcal{P}(\mathbb{R})'$ defined by $\phi(p) = p''(7)$.

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Suppose $\phi \in \mathcal{P}(\mathbb{R})'$ defined by $\phi(p) = p''(7)$.

Describe $T'(\phi)$ on $\mathcal{P}(\mathbb{R})$.

Examples and Exercises

Suppose V is finite-dimensional and U is a subspace of V such that $U \neq V$. Prove that there exists $\phi \in V'$ such that $\phi(u) = 0$ for every $u \in U$ but $\phi \neq 0$.

Examples and Exercises

Suppose V is finite dimensional and $v \in V$ with $v \neq 0$. Prove that there exists $\phi \in V'$ such that $\phi(v) = 1$.

Annihilator

Def'n:

For $U \subset V$ the **annihilator** of U is denoted U^0 and is defined to be

$$U^0 = \{\phi \in V' : \phi(u) = 0 \text{ for all } u \in U\}.$$

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$\phi \in \mathcal{P}(\mathbb{R})'$ defined as $\phi(p) = p'(0)$ is in U^0

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Prop'n:

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Using definitions, replace $\dim \operatorname{null} i'$ with $\dim U^0$.
Need: $\dim \operatorname{range} i' = \dim U$...
$\operatorname{range} i' = U'$ because every linear functional of U can be extended to V

Null space of T'

Prop'n:

Suppose V and W are finite dimensional $T \in \mathcal{L}(V, W)$. Then

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Proof notes.

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part 2: Notice $\dim(\operatorname{range} T)^0 = \dim W - \dim \operatorname{range} T$.

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$(\text{range } T)^0 = 0 \Leftrightarrow \text{null } T' = 0$

Similar Results

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Prop'n [Axl14]:

Suppose V and W are finite dimensional $T \in \mathcal{L}(V, W)$. Then T is injective if and only if T' is surjective.

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Def'n:

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- For $T \in \mathcal{L}(V, W)$, $\mathcal{M}(T') = (\mathcal{M}(T))^t$.

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From definition, $T'(\psi_j) = \sum_{r=1}^{n} C_{r,j}\phi_r$ for some bases of W', V'.

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$$(\psi_j \circ T)(v_k) = \sum_{r=1}^{n} C_{r,j}\phi_r(v_k) = C_{k,j}.$$

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$$(\psi_j \circ T)(v_k) = \sum_{r=1}^{n} C_{r,j}\phi_r(v_k) = C_{k,j}.$$

On the other hand, we also have

$$\begin{aligned}(\psi_j \circ T)(v_k) &= \psi_j(Tv_k) \\ &= \psi_j(\sum_{r=1}^{m} A_{r,k}w_r) \\ &= \sum_{r=1}^{m} A_{r,k}\psi_j(w_r)\end{aligned}$$

Matrix Rank

Def'n:

Suppose A is an $m \times n$ matrix with entries in $\mathbb{F}$.

- The ***row rank*** is the dimension of the span of the rows of A.
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Examples:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 8 & 7 & 0 & 6 \end{pmatrix}$$

Range and Column Rank

Prop'n:

Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $\dim \operatorname{range} T$ is equal to the column rank of $\mathcal{M}(T)$.

Proof sketch. Pick $v_1, \ldots, v_n$ to be a basis of V.

$\mathcal{M} : \operatorname{span}(Tv_1, \ldots, Tv_n) \to \operatorname{span}(\mathcal{M}(Tv_1), \ldots, \mathcal{M}(Tv_n))$ is an isomorphism.

$\operatorname{range} T = \operatorname{span}(Tv_1, \ldots, Tv_n)$ and
$\dim \operatorname{span}(\mathcal{M}(Tv_1), \ldots, \mathcal{M}(Tv_n)) = \text{column rank}$

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$$\begin{aligned}
\text{column rank of } A &= \text{column rank of } \mathcal{M}(T) \\
&= \dim \operatorname{range} T \\
&= \dim \operatorname{range} T' \\
&= \text{column rank of } \mathcal{M}(T') \\
&= \text{column rank of } A^t \\
&= \text{row rank of } A
\end{aligned}$$

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$\square$

Matrix Rank

Def'n:

We then say the **rank** of a matrix is the column rank (or equivalently the row rank).

References

[Axl14] Sheldon Axler.
Linear Algebra Done Right.
Undergraduate Texts in Mathematics. Springer Cham, 2014.